

Riley James Link

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RESEARCH INTERESTS

I am a first year PhD student in Applied Mathematics at Cornell University with research interests in geometry, optimization, control theory, and robotics, with a focus on developing mathematical frameworks to solve complex real-world problems.

EDUCATION

Cornell University - Ithaca, New York

May 2030

Doctorate of Philosophy (PhD) in Applied Mathematics

- *Relevant Coursework:* Bayesian Estimation & Stochastic Optimization, Probability 1, Mathematical Programming I, Probability 2, Continuous Optimization

Creighton University - Omaha, Nebraska

May 2025

Bachelor of Science in Mathematics

GPA 4.0

- *Relevant Coursework:* Data Structures, Computational Physics, Classical Mechanics, Statistical Modeling, Electricity & Magnetism, Real Analysis, Measure Theory, Partial Differential Equations

RESEARCH EXPERIENCES

Position, Navigation and Timing: Algorithms and Analysis Group – MITRE

May 2024 – August 2024

Signal Processing Engineering Intern

- Tasked with redefining performance bounds on satellite based geolocation
- Used signal processing TDOA and FDOA geolocation techniques
- Completed statistical analysis using Cramer-Rao Lower Bounds
- Wrote python code base ready to be integrated into existing repositories
- Updated existing MITRE technical report to include new performance bounds

DRUMS Research Experience for Undergraduates – North Carolina State University

May 2023 – May 2024

Researcher

- Participated in summer REU funded by the NSF and NSA
- Continued work into fall and spring semesters with weekly meetings to discuss progress
- Completed a group project on *A Framework for Approximating Perturbed Optimal Control Problems*
- Co-authored and published results in the SIAM Undergraduate Research Online (SIURO)
 - DOI: [10.1137/24S1686000](https://doi.org/10.1137/24S1686000)
- Poster presentation at the NCSU Summer Undergraduate Research & Creativity Symposium
- Individual Contributed Undergraduate Talk at the *Iowa MAA Fall 2023 Section Meeting*
- Co-presented research at the *2024 Joint Math Meetings Undergraduate Research Poster Session*
- Presented research at the *2025 Joint Math Meetings Contributed Session on Research by Undergraduates*

PROGRAMMING SKILLS

Languages

- MATLAB, Python, Java, R, C++

Tools

- MATLAB Optimization Toolbox, Data Structures, Object-Oriented Programming, Numpy, Pandas, Matplotlib, Scipy, Jupiter Notebooks, Latex, Beamer, Git

Activities

- Wrote over 3000 lines of code in MATLAB for summer 2023 project dealing with trajectory optimization
- Worked on projects that involved Convolutional Neural Networks (CNN) in Machine Learning course
- Worked on image deconvolution project that also included a CNN in Computational Physics course
- Implemented the Mastermind Donald Knuth Five-guess algorithm in C++

EMPLOYMENT

Department of Mathematics – Cornell University

August 2025 – Present

Teaching Assistant and Grader – MTH 1920: Multivariable Calculus for Engineers

- Lead weekly discussion sections reinforcing core topics
- Held regular office hours to provide one-on-one and small-group tutoring
- Grade students weekly homework, providing constructive feedback to help improve students understanding

Department of Mathematics – Creighton University

August 2022 – May 2025

Teaching Assistant and Grader – MTH 350: Applied Linear Algebra and Differential Equations

- Host one-and-a-half hour long sessions twice a week that consist of open office hours in which students ask questions concerning material or homework
- Communicate with professor concerning students struggling with material or specific questions
- Grade students weekly homework according to guidelines given by professor

New Horizon Tutoring - North Sioux City, SD

May 2022 – February 2023

Mathematics Online Tutor

- Design sufficient lesson plans for each tutoring session beforehand including examining textbook passages, preparing questions to engage students, and creating practice problems
- Tutor high school to early college students in one-on-one, online weekly sessions on math courses such as Algebra 1, Geometry, Algebra 2, Precalculus, Calculus 1, and Calculus 2
- Earned the *Rookie Tutor of the Year* award in 2022

Department of Chemistry – Creighton University

August 2021 – May 2025

Work Study

- Assist the department chair, laboratory manager, and any professor with the task at hand
- Provide clerical and office support, including delivering confidential materials to professors, helping any students with general questions about the department, selling chemistry lab manuals and books, and other supporting tasks
- Aid in the preparation of laboratory's each week along with any other duties needed in the stockroom

Department of Residential Life – Creighton University

August 2023 – May 2025

Honors Lead Resident Advisor

- Develop a positive living environment for residents by creating mediation and conflict resolution skills
- Implement and enforce all College and Campus Living policies, promoting academic excellence, addressing student needs, and encouraging involvement
- Collaborate with orientation staff on first year and transfer student opening week activities

ACTIVITIES, HONORS, AND LEADERSHIP EXPERIENCE

Creighton University

- College of Arts & Sciences Dean's Award for Distinguished Excellence, 1 of 4 graduating seniors honored
- Bernard Portz, S.J. Award for Excellence in Mathematics
- **National Barry Goldwater Scholarship Recipient**
- Phi Beta Kappa Membership
- Creighton Honors Program Membership
- Pi Mu Epsilon Mathematics Honor Society Membership
- Alpha Sigma Nu Jesuit Honor Society Membership
- DREAM Scholars Volunteer (August 2023 – Present)
- Chosen for 2023/24 'Consider Initiative', delivered public talk *Math: Our Silent Ally (and Why it isn't Terrifying)*
- Chess Club, Tennis Club, Badminton Club, and Omaha and Council Bluffs Pickleball Member
- Wrote and directed a short film titled *Perspective*: <https://youtu.be/efpOr8c10rQ>

Weiss High School

- Valedictorian of a graduating class of 480 & AP Scholar with Distinction